L'ORÉAL SUSTAINABILITY CHALLENGE 2026 · OPTION 02: LONGEVITY & FUTURE AGING

SKINNOVA · Climate-Adaptive Skin Longevity Platform for Indian GenZ

RESEARCH & INNOVATION TRACK

Target Brands: L'Oréal Paris · CeraVe · Garnier

Zero-Backend PWA · 100% On-Device CV · Live Open-Meteo

01 · SCAN

On-Device Canvas CV

Forehead & bilateral malar patches in CIELAB D65; Laplacian micro-roughness. Photo never leaves phone.

02 · CLIMATE

Live Hyper-Local Telemetry

Open-Meteo API ingestion (UV, PM2.5, Temp, RH) across 8 Indian cities; silent sample fallback.

03 · FUSION

CSSI & Resilience Score

Transparent 0-100 heuristic referencing WHO 2021 & CPCB NAQI; self-anchored delta calibration.

04 · ROUTINE

Adaptive Minimalist Protocol

CSSI × Signal matrix; fluid SPF 50+, 4% Niacinamide, Ceramides, & 'Skip-Today' bypass state.

THE PROBLEM · INDIAN GENZ CLIMATE TRILEMMA

Environmental Stress Accelerates Invisible Aging

- **PM2.5 / Winter Smog (>180 µg/m³):** Particulate-bound PAHs activate AhR receptor, triggering ROS & MMP-1 collagenase degradation.
- **Extreme Solar UV (Index 11-12+):** Tropical solar irradiance drives mitochondrial DNA damage & rapid squalene peroxidation.
- **Tropical Heat & Humidity (>80% RH):** Promotes follicular occlusion, oxidized sebum plugs, and barrier breakdown from 10-step over-layering.
- **Melanin-Rich Vulnerability:** Micro-inflammation induces stubborn Post-Inflammatory Hyperpigmentation (PIH) in Fitzpatrick phototypes III-VI.

FUSION ENGINE · CSSI & SKIN RESILIENCE

Transparent Scientific Heuristic (0–100)

```
CSSI = 0.35·S(UV) + 0.35·S(PM2.5) + 0.20·S(RH_Heat) +
0.10·S(Temp)
Resilience Score = clamp(100 - (0.55·CSSI + 0.45·VisibleStress),
5, 98)
```

Clinical Validation Plan (Non-Invasive Gold Standards):

- **Corneometer® CM 825:** Validates predicted barrier hydration vs stratum corneum capacitance (target r ≥ 0.78).
- **Tewameter® TM 300:** Inverse correlation (r ≤ -0.72) against vapor-flux Trans-Epidermal Water Loss (TEWL).
- **Mexameter® MX 18:** Spectrophotometric calibration of Δa* against Melanin & Erythema Indices in Fitzpatrick III–VI.

SUSTAINABILITY · BIODIVERSITY ACT & CIRCULARITY

Ethical Sourcing & Anti-Overconsumption

- **India Biodiversity Act Compliance:** Access & Benefit Sharing (ABS) partnership with tribal farmer cooperatives in Madhya Pradesh for wild Amla.
- **Circular Packaging:** Refillable 100% PCR aluminum cartridge with compostable PHA inner pods; reduces virgin plastics by 78%.
- **'Skip-Today' Anti-Overconsumption:** Algorithmically bypasses heavy actives on low-stress days (target ~4g CO_2e avoided per skipped step).
- **Edge Compute Efficiency:** On-device processing uses ~0.002 Wh/scan vs ~0.08 Wh for cloud GPU server API roundtrips.

CORE DIFFERENTIATOR · ETHICAL ON-DEVICE AI

Fairness-by-Design & Zero-Leakage Privacy

- **100% Client-Side Privacy:** Zero facial images uploaded or stored on cloud servers. Instant sub-1.5s load via QR code with zero user login.
- **Baseline-Relative Calibration:** Signals evaluated against individual calibrated baseline (ΔL*, Δa*, ΔLaplacian), mathematically preventing pale-skin bias across Indian skin.
- **Lighting Quality Gate:** Active checks on mean L*, dynamic range, and cheek symmetry block corrupted scans with friendly retake cues.

PAIRED FORMULATION · CONCEPT #SK-26-IND

Bio-Adaptive Antioxidant & Ceramide Gel-Cream

Sensory: Water-break micro-droplet gel-cream; zero pore-clogging film under 80%+ humidity.

- **2% Phyllanthus Emblica (Amla):** ABS-certified polyphenol complex suppressing PM2.5-induced collagenase (MMP-1).
- **4% Pharmaceutical Niacinamide:** Regulates sebum flux in tropical heat & accelerates epidermal ceramide synthesis.
- **1.5% Bio-Identical Ceramides:** Yeast-derived NP, AP, EOP rebuilding intercellular lipid lamellar bilayers.
- **1% Stabilized Ethyl Ascorbic Acid:** High thermal stability (to 45°C); combats melanin clustering without phototoxicity.

GO-TO-MARKET · BRAND PORTFOLIO INTEGRATION

Radical Transparency for Indian GenZ

- **Consumer Mindset:** 18-25 urbanites fatigued by multi-step hype; seeking honest barrier science, zero white cast, and proven climate defense.
- **L'Oréal Synergy:** CeraVe India (clinical barrier diagnostics), L'Oréal Paris (youth longevity), or Garnier (biodiverse Indian botanicals).
- **Campus & Retail QR Mirrors:** Zero-install QR web app at university tech fests, Nykaa, and Tira beauty counters; loads in < 1.5 seconds.
- **Core Brand Message:** 'Protect the barrier, skip the excess. Climate longevity engineered for Indian skin.'

Concept proposal; metrics are targets pending validation. | Live Web App: https://koushik1914.github.io/skinnova/ | **Koushik Voota | Amrita Vishwa Vidyapeetham**